CAPTURE THE FLAG (CTF) MACHINE DEVELOPMENT REPORT

WestWild CTF Machine Development Using Kali Linux

Submitted By
Name: Tasleem Shaikh
SRN-R25DE191
Program: MCA
Semester: II
Course: Cyber Security Project

Abstract

This report documents the development, exploitation, and customization of the WestWild Capture The Flag machine using Kali Linux. It covers reconnaissance, enumeration, SMB access, credential discovery, SSH login, privilege escalation, website customization, user management, hostname configuration, and custom flag creation.

Objective

• Discover the target
• Enumerate services
• Access SMB shares
• Gain SSH access
• Escalate privileges
• Customize the machine
• Document the complete process.

Machine Specifications

Target: WestWild
Attacker: Kali Linux
OS: Ubuntu 14.04.6 LTS
Web Server: Apache2
Network: NAT
Services: SSH, HTTP, SMB

Operating System Used

Target: Ubuntu 14.04.6 LTS
Attacker: Kali Linux

Network Configuration

Both VMs were configured in VMware NAT mode to allow isolated communication.

1. VMware Configuration

Tool Used: VMware Workstation

Purpose: Complete the corresponding CTF task.

Commands Used:

GUI only

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

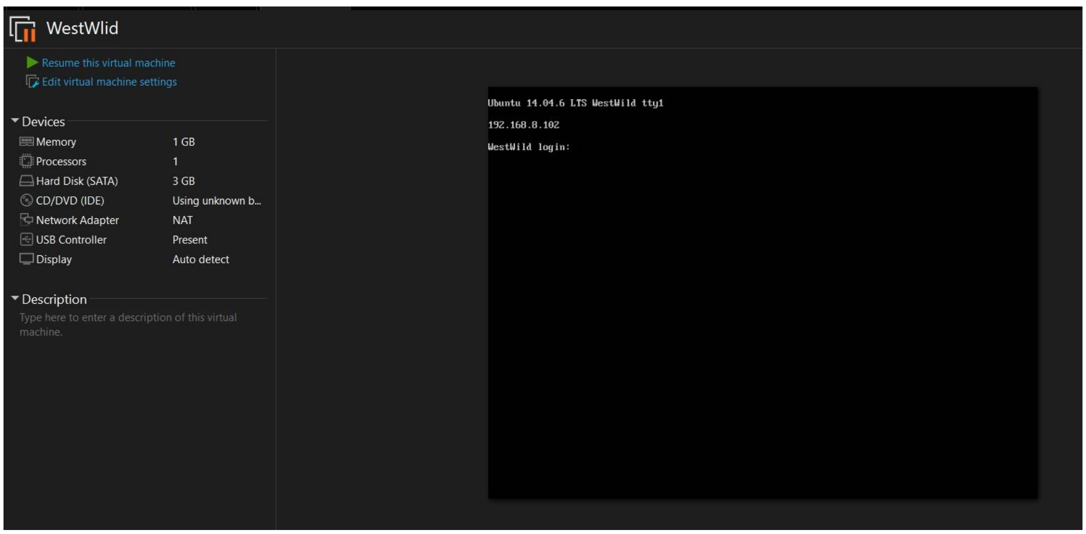


Figure: 1. VMware Configuration

2. Host Discovery

Tool Used: Netdiscover

Purpose: Complete the corresponding CTF task.

Commands Used:

netdiscover

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
4 Captured ARP Req/Rep packets, from 4 hosts.   Total size: 240
_____________________________________________________________________________
 IP            At MAC Address     Count     Len  MAC Vendor / Hostname
_____________________________________________________________________________
192.168.17.1   00:50:56:c0:00:08      1      60  VMware, Inc.
192.168.17.2   00:50:56:f6:7a:6f      1      60  VMware, Inc.
192.168.17.135 00:0c:29:bd:c6:32      1      60  VMware, Inc.
192.168.17.254 00:50:56:f8:51:58      1      60  VMware, Inc.
```

Figure: 2. Host Discovery

3. Port Enumeration

Tool Used: Nmap

Purpose: Complete the corresponding CTF task.

Commands Used:

nmap -A 192.168.228.135

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
┌──(root㉿kali)-[/home/kali]
└─# nmap -A 192.168.17.135
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-04 09:42 -0400
Nmap scan report for 192.168.17.135
Host is up (0.00089s latency).
Not shown: 996 closed tcp ports (reset)
PORT    STATE SERVICE     VERSION
22/tcp  open  ssh         OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 6f:ee:95:91:9c:62:b2:14:cd:63:0a:3e:f8:10:9e:da (DSA)
|   2048 10:45:94:fe:a7:2f:02:8a:9b:21:1a:31:c5:03:30:48 (RSA)
|   256 97:94:17:86:18:e2:8e:7a:73:8e:41:20:76:ba:51:73 (ECDSA)
|_  256 23:81:c7:76:bb:37:78:ee:3b:73:e2:55:ad:81:32:72 (ED25519)
80/tcp  open  http        Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
|_http-title: Site doesn't have a title (text/html).
139/tcp open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open  netbios-ssn Samba smbd 4.3.11-Ubuntu (workgroup: WORKGROUP)
MAC Address: 00:0C:29:BD:C6:32 (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14
Network Distance: 1 hop
Service Info: Host: WESTWILD; OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Figure: 3. Port Enumeration

4. SMB Enumeration

Tool Used: SMBClient

Purpose: Complete the corresponding CTF task.

Commands Used:

smbclient -L //192.168.228.135

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

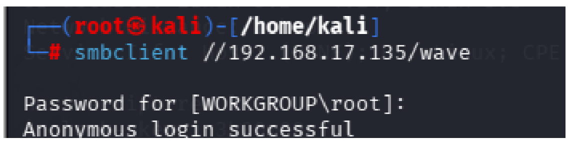


Figure: 4. SMB Enumeration

5. Download Files

Tool Used: SMBClient

Purpose: Complete the corresponding CTF task.

Commands Used:

smbclient //192.168.228.135/wave
ls
get FLAG1.txt
get message_from_aveng.txt

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

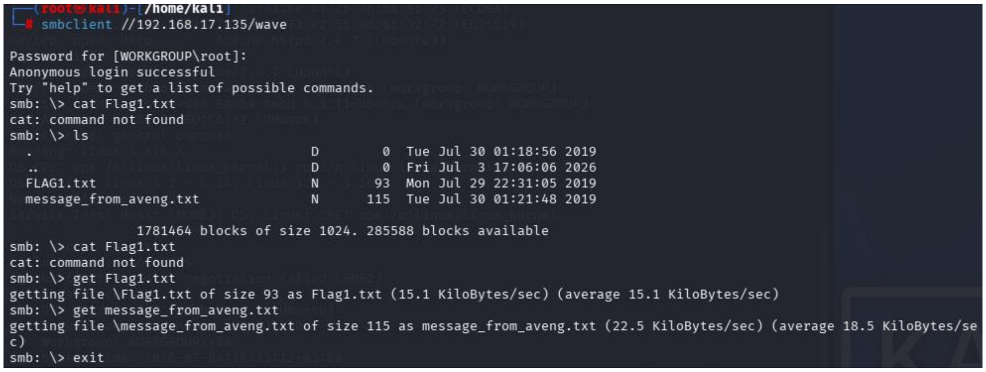


Figure: 5. Download Files

6. Read Files

Tool Used: cat

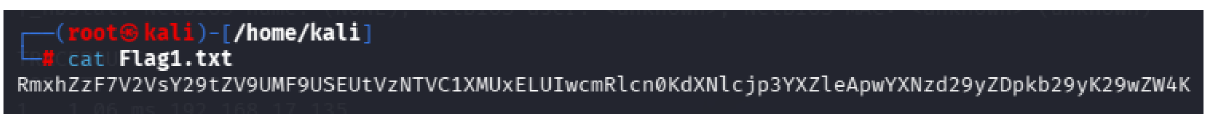
Purpose: Complete the corresponding CTF task.

Commands Used:

cat FLAG1.txt
cat message_from_aveng.txt

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.



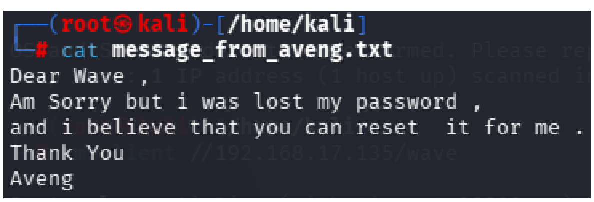


Figure: 6. Read Files

7. Base64 Decode

Tool Used: base64

Purpose: Complete the corresponding CTF task.

Commands Used:

echo "ENCODED_STRING" | base64 -d

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

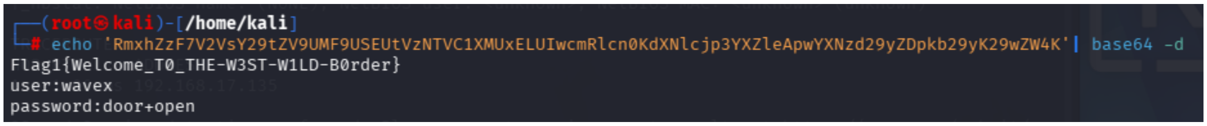


Figure: 7. Base64 Decode

8. SSH Login

Tool Used: SSH

Purpose: Complete the corresponding CTF task.

Commands Used:

ssh wavex@192.168.228.135

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

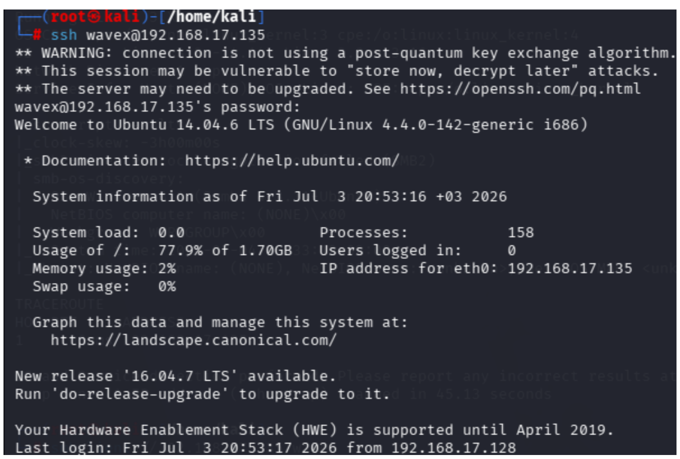


Figure: 8. SSH Login

9. Writable Directories

Tool Used: find

Purpose: Complete the corresponding CTF task.

Commands Used:

find / -writable -type d 2>/dev/null

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
wavex@WestWild:~$ find / -writable -type d 2>/dev/null
/sys/fs/cgroup/systemd/user/1001.user/2.session
/usr/share/av/westsidesecret
/home/wavex
/home/wavex/.cache
/home/wavex/wave
/var/lib/php5
/var/spool/samba
/var/crash
/var/tmp
/proc/1875/task/1875/fd
/proc/1875/fd
/proc/1875/map_files
/run/user/1001
/run/shm
/run/lock
/tmp
```

Figure: 9. Writable Directories

10. Credential Discovery

Tool Used: cat

Purpose: Complete the corresponding CTF task.

Commands Used:

cd /usr/share/av/westsidesecret
ls
cat ififorget.sh

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
wavex@WestWild:~$ cd /usr/share/av/westsidesecret
wavex@WestWild:/usr/share/av/westsidesecret$ ls
ififoregt.sh
wavex@WestWild:/usr/share/av/westsidesecret$ /usr/share/av/westsidesecret$ cat ififoregt.sh
-bash: /usr/share/av/westsidesecret$: No such file or directory
wavex@WestWild:/usr/share/av/westsidesecret$ /usr/share/av/westsidesecret$ cat ififoregt.sh #!/bin/bash
-bash: /usr/share/av/westsidesecret$: No such file or directory
wavex@WestWild:/usr/share/av/westsidesecret$  cat ififoregt.sh
 #!/bin/bash
 figlet "if i foregt so this my way"
 echo "user:aveng"
 echo "password:kaizen+80"
```

Figure: 10. Credential Discovery

11. User Switch

Tool Used: su

Purpose: Complete the corresponding CTF task.

Commands Used:

su aveng

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

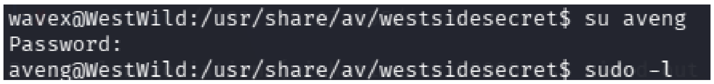


Figure: 11. User Switch

12. Sudo Privileges

Tool Used: sudo

Purpose: Complete the corresponding CTF task.

Commands Used:

sudo -l

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

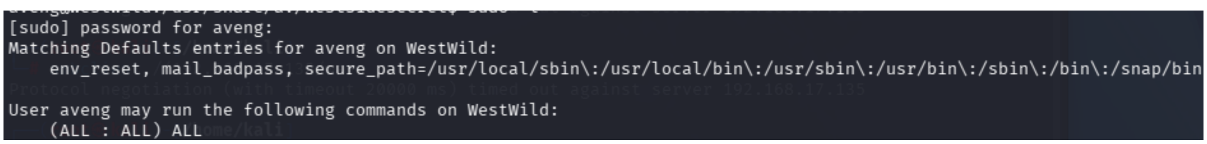


Figure: 12. Sudo Privileges

13. Root Access

Tool Used: sudo

Purpose: Complete the corresponding CTF task.

Commands Used:

sudo su

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

```
aveng@WestWild:/usr/share/av/westsidesecret$ sudo su
root@WestWild:/usr/share/av/westsidesecret# cd
```

Figure: 13. Root Access

14. Root Flag

Tool Used: cat

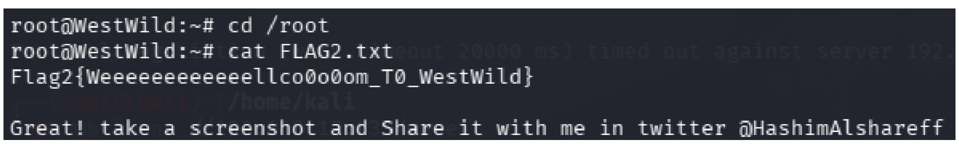
Purpose: Complete the corresponding CTF task.

Commands Used:

> ***cd /root***
> ***ls***
> ***cat FLAG2.txt***

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.



```
root@WestWild:~# cd /root
root@WestWild:~# cat FLAG2.txt
Flag2{Weeeeeeeeeeeellco0o0om_T0_WestWild}

Great! take a screenshot and Share it with me in twitter @HashimAlshareff
```

Figure: 14. Root Flag

15. Apache Directory

Tool Used: Apache2

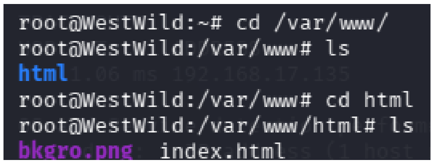
Purpose: Complete the corresponding CTF task.

Commands Used:

> ***cd /var/www***
> ***cd html***
> ***ls***

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.



```
root@WestWild:~# cd /var/www/
root@WestWild:/var/www# ls
html
root@WestWild:/var/www# cd html
root@WestWild:/var/www/html# ls
bkgro.png  index.html
```

Figure: 15. Apache Directory

16. Website Editing

Tool Used: Nano

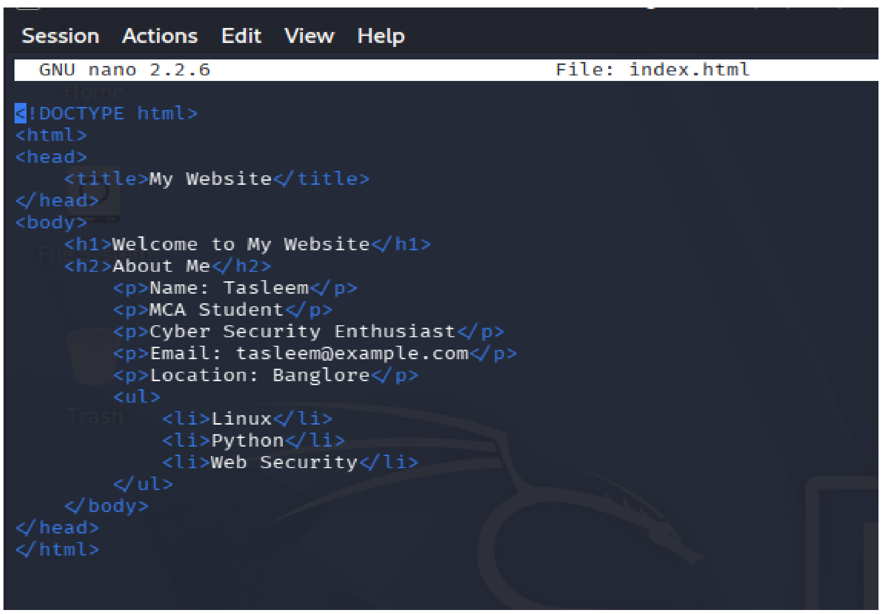
Purpose: Complete the corresponding CTF task.

Commands Used:

nano index.html

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.



```
Session  Actions  Edit  View  Help
  GNU nano 2.2.6                          File: index.html

<!DOCTYPE html>
<html>
<head>
    <title>My Website</title>
</head>
<body>
    <h1>Welcome to My Website</h1>
    <h2>About Me</h2>
        <p>Name: Tasleem</p>
        <p>MCA Student</p>
        <p>Cyber Security Enthusiast</p>
        <p>Email: tasleem@example.com</p>
        <p>Location: Banglore</p>
        <ul>
            <li>Linux</li>
            <li>Python</li>
            <li>Web Security</li>
        </ul>
    </body>
</head>
</html>
```

Figure: 16. Website Editing

17. Apache Service

Tool Used: service

Purpose: Complete the corresponding CTF task.

Commands Used:

service apache2 start
service apache2 stop

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

Figure: 17. Apache Service

18. Website Verification

Tool Used: Browser

Purpose: Complete the corresponding CTF task.

Commands Used:

http://192.168.228.135

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

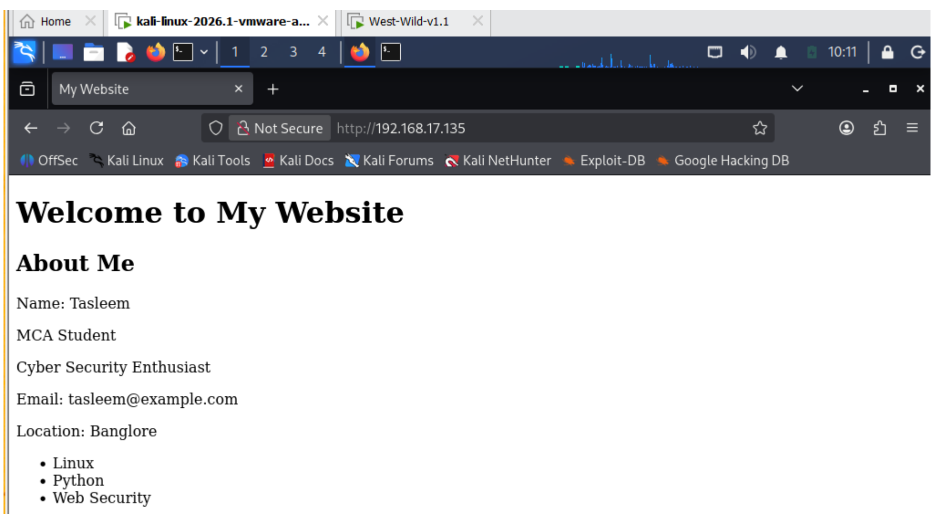


Figure: 18. Website Verification

19. User Creation

Tool Used: useradd/passwd

Purpose: Complete the corresponding CTF task.

Commands Used:

> ***useradd user1***
> ***passwd user1***
> ***useradd user2***
> ***passwd user2***

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

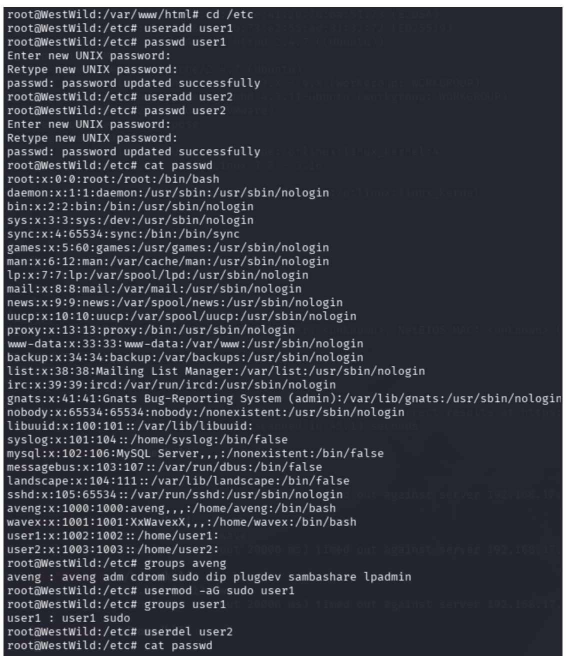


Figure: 19. User Creation

20. Group Management

Tool Used: groups/usermod

Purpose: Complete the corresponding CTF task.

Commands Used:

> ***groups aveng***
> ***usermod -aG sudo user1***
> ***groups user1***

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

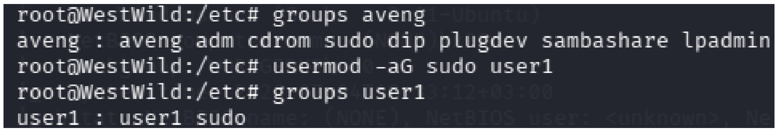


Figure: 20. Group Management

21. User Deletion

Tool Used: userdel

Purpose: Complete the corresponding CTF task.

Commands Used:

userdel user2

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

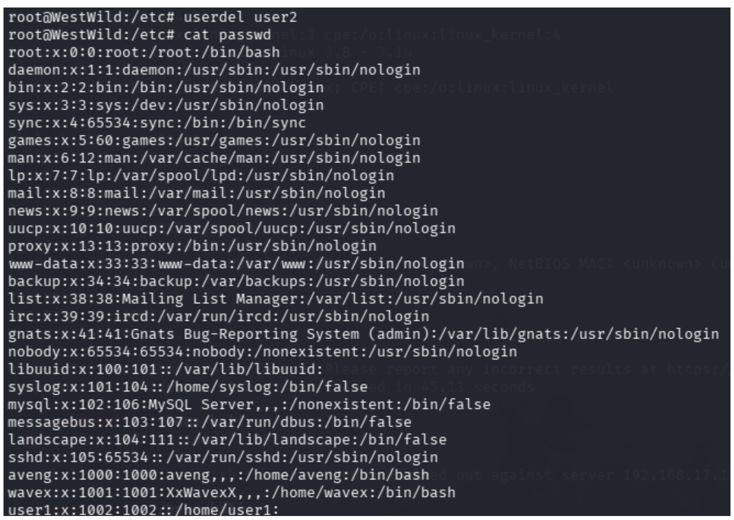


Figure: 21. User Deletion

22. Hostname Configuration

Tool Used: hostname/nano

Purpose: Complete the corresponding CTF task.

Commands Used:

hostname
nano /etc/hostname

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

[INSERT SCREENSHOT HERE]

Figure: 22. Hostname Configuration

23. Hostname Verification

Tool Used: hostname

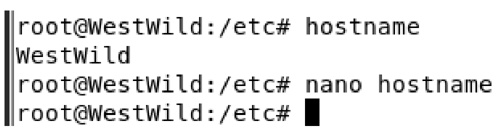
Purpose: Complete the corresponding CTF task.

Commands Used:

hostname

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.



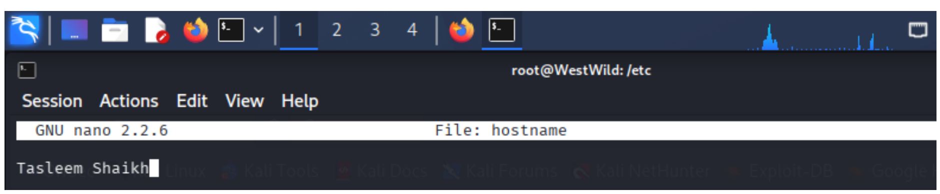


Figure: 23. Hostname Verification

24. Login Screen Verification

Tool Used: Linux Console

Purpose: Complete the corresponding CTF task.

Commands Used:

Reboot and verify hostname Tasleem

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

25. Custom Flag

Tool Used: nano/cat

Purpose: Complete the corresponding CTF task.

Commands Used:

> ***nano FLAG2.txt***
> ***cat FLAG2.txt***

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

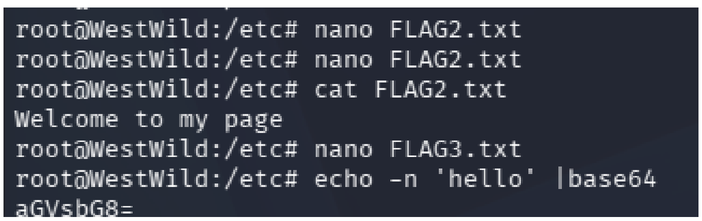


Figure: 25. Custom Flag

26. Base64 Encoding

Tool Used: base64

Purpose: Complete the corresponding CTF task.

Commands Used:

> ***echo -n 'hello' | base64***

Explanation: Perform the activity shown above and verify the expected result.

Observation: The task completed successfully.

Tools Summary

VMware, Netdiscover, Nmap, SMBClient, SSH, cat, base64, find, su, sudo, Apache2, Nano, useradd, passwd, usermod, userdel, hostname.

Testing Procedure

Perform reconnaissance, enumeration, exploitation, privilege escalation, machine customization, website deployment, user management, hostname verification, and final validation.

Conclusion

The WestWild CTF machine was successfully exploited and customized. The exercise demonstrated practical penetration testing and Linux administration skills, including reconnaissance, enumeration, privilege escalation, Apache configuration, user management, and system customization.